Yufan Zhou

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(CV updated on Oct 18, 2026)

Education

- **Nanjing University** — **Nanjing**
  Ph.D. in Astronomy and Astrophysics, Advisor: Prof. Zhiyuan Li — *2022–2027*
- **Nanjing University** — **Nanjing**
  B.S. in Astronomy and Astrophysics — *2018–2022*

Research Interests

- N-body simulations and the application of AI in astronomy
- Mergers of galaxies and supermassive black holes in the Local Group
- Formation and Destruction of Stellar Bars and Spiral Arms Caused by Galactic collisions
- Orbital Evolution of Small Bodies in the Solar System

Publications

4 first/corresponding author (+1 submitted), 1 co-author

First/corresponding Author:

F5. **Zhou, Yufan**; Zhang, Tianchi; Liao, Shihong; Li, Zhiyuan; Wang, Lan; Benisty, David. 2026, submitted to ApJ. "Not So Small: The Influence of Hubble Expansion on the Future Merger of the Milky Way and Andromeda"

F4. **Zhou, Yufan**; Li, Zhiyuan; Jiménez-Arranz, Óscar; Roca-Fàbrega, Santi. 2026, ApJ, 1002, 44. "Survival or Destruction: Effects of Spheroidal Satellite Collisions on Bars in Milky Way-Like Galaxies"

F3. **Zhou, Yufan**; Li, Zhiyuan; Li, Hui. 2025, ApJ, 988, 205. "N-body Simulations of Galaxy Bars Generated by Satellite Collisions: Effects of the Impact Geometry"

F2. **Zhou, Yufan**; Li, Zhiyuan; Li, Hailiang; Zhou, Liyong. 2025, MNRAS, 538, 258. "Prediction for close approaches with terrestrial planets of asteroids from the main belt"

F1. **Zhou, Yufan**; Li, Hailiang; Li, Zhiyuan; Zhou, Liyong. 2024, MNRAS, 532, L7. "MARTIANS (MARs2020, TIANwen and So on) would see more potentially hazardous asteroids than Earthlings"

Co-Author:

C1. Li, Zongnan; Wang, Sumin; Su, Zhao; **Zhou, Yufan**; et al. 2025, ApJ, 983, 182. "Ring or no ring – Revisiting the Multiphase Nuclear Environment in M31"

Accepted Proposals

- "Hunting for Near-Mars Asteroids with JST", 2025, Co-PI, Observatorio Astrofísico de Javalambre: JST250-JPCam, 5.41h
- "Illuminating the Outburst: A SITELLE View of M31's Turbulent Past", 2025, Co-I, Canada France Hawaii Telescope: SITELLE, 6h
- "Smoking Gun of A kpc-scale Outflow in the Disk of M31?", 2023, Co-I, Canada France Hawaii Telescope: SITELLE, 5.3h

- "A Triple Galaxy System Caught in the Act of Merging", 2023, Co-I, Chandra: ACIS, 60ks

Technical skills

- **Programming Languages:** Python, C, C++, Fortran
- **Simulation Codes:** GADGET, GIZMO, KETJU, Mercury6, PeTar
- **Parallel Computing:** MPI, OpenMP, CUDA
- **Analysis and Writing:** Matlab, LaTeX

Conferences & Seminars

- **Invited Seminar** at NAOC, Beijing, China, Oct. 2026
- **Talk** at Symposium on the Dwarf Galaxies, Suzhou, China, Apr. 2026
- **Invited Seminar** at International Teeminar Online Meeting (Heidelberg & Beijing), Online, Mar. 2026
- **Talk** at Cosmic Collisions Conference (NYU Abu Dhabi), Abu Dhabi, The United Arab Emirates, Jan. 2026
- **Talk** at Seminar on Galaxy Mergers and Interactions in the Age of Space Telescopes, Suzhou, China, Jun. 2025
- **Talk** at The 6th Youth Planetary Forum, Huangshan, China, Mar. 2025
- **Talk** at Frontier Symposium on Planetary Science and Deep Space Exploration of the CAS, Xiamen, China, Dec. 2024
- **Talk** at Scientific Symposium on the Andromeda Galaxy M31, Suzhou, China, Oct. 2024
- **Talk** at National Planetary Science Conference, Nanjing, China, Oct. 2024
- **Talk** at The 45th COSPAR Scientific Assembly, Busan, Korea, Jul. 2024
- **Talk** at The 3rd China Space Science Assembly, Huzhou, China, Oct. 2023
- **Talk** at Annual Meeting of the Chinese Astronomical Society, Weihai, China, Aug. 2023

Honors and Awards

- The First Award of Doctoral Student Talent Scholarship in Nanjing University, 2025
- Excellent Postgraduate Student of Nanjing University, 2025

Outreach

- **Popular science article published in a Chinese magazine** "Travel on a spider - Use spider pulsars for interstellar travel". Amateur Astronomer, 2025, 04: 56-59 (based on arXiv:2411.05038)
- **Volunteer** at the Zuodijiang Observatory Open Day (Apr. 2023; Apr. 2024; Apr. 2025; Nov. 2025)

References

- **Dr. Zhiyuan Li**, Professor, Nanjing University, lizy@nju.edu.cn
- **Dr. Shihong Liao**, Associate Researcher, National Astronomical Observatories (CAS), shliao@nao.cas.cn
- **Dr. Santi Roca-Fàbrega**, Researcher, Lund University, santi.roca_fabrega@fysik.lu.se